# COMSATS University Islamabad
## Department of Computer Science

### Course Information

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|---|---|
| Course Code: **CSC354** | Course Title: **Machine Learning** |
| Credit Hours: **3(3,0)** | Lecture Hours/Week: **3** |
| Lab Hours/Week: **0** | Pre-Requisites: **None** |

### Text and Reference Books

**Textbooks:**

1. Introduction to Machine Learning, Ethem Alpaydin, MIT Press, 2010.
2. Machine Learning, Tom, M., McGraw Hill, 1997.

**Reference Books:**

1. Hands on Machine Learning with Scikit-Learn and TensorFlow, Aurelien Geron, O`Reilly Media, 2017.
2. Deep Learning with PyTorch – Essential Excerpts, Eli Stevens, Luca Antiga, Thomas Viehmann, Manning Publications, 2009.
3. Pattern Recognition and Machine Learning, Bishop, C., Springer-Verlag, 2007.

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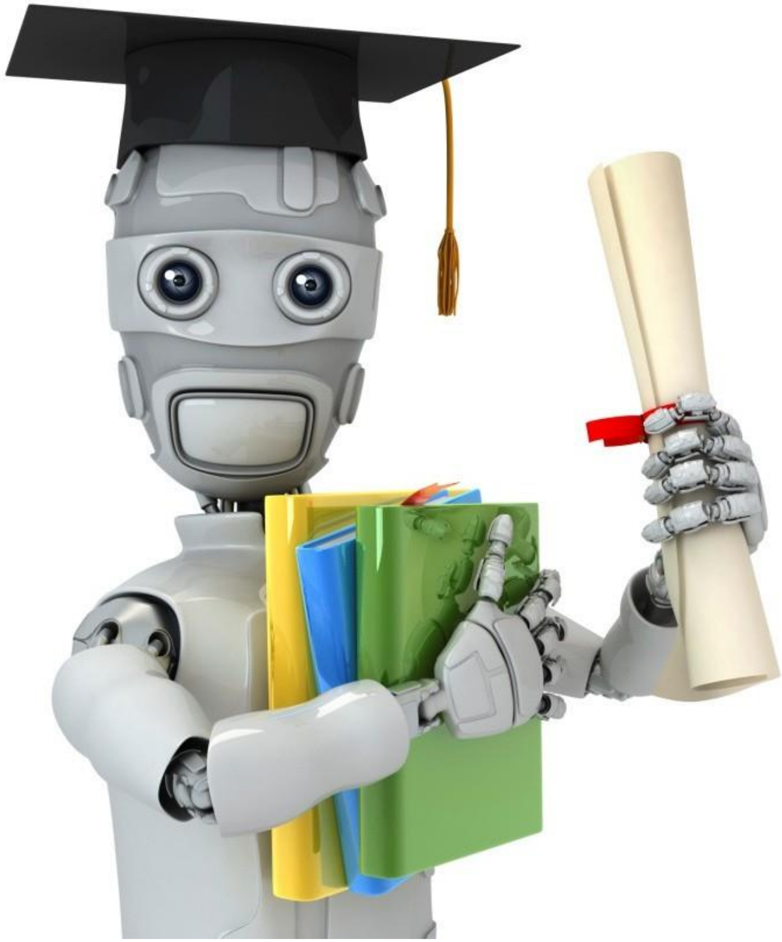
- Zero tolerance for any misconduct and misbehavior
- Zero tolerance for cheating, if student found with any cheating material in quiz/exam will be marked zero
- Copied assignment will be marked zero
- Cross talk with class-fellows will result in strict disciplinary action
- Be in class on time, students who arrive after 5 min will be marked absent
- Submit assignments on portal on time, late assignment wouldn't be considered, and anything send via (WhatsApp etc) wouldn't be considered
- Emails or any message should be sent within working hours rest would not be entertained send email on **(zeenatzulfiqar@cuiatd.edu.pk)**



Machine Learning

Reinforcement Learning:

- Hidden Markov Model
- Monte Carlo
- Q Learning

# Hidden Markov Model

- A **Hidden Markov Model (HMM)** is a **probabilistic model** used to analyze **sequential or time-series data**. It is useful when the **actual states of a system cannot be observed directly**, but we can observe their effects or outputs.
- The word **"Hidden"** means the real state is unknown, while **"Markov"** means the next state depends only on the current state, not on the entire history.
- **An HMM assumes that:**
- The system moves from one hidden state to another over time.
- Each hidden state generates an observable output according to a probability.
- By observing the outputs, we can estimate the most likely hidden states.

# Components of a Hidden Markov Model

- A Hidden Markov Model consists of **five main components**.
- **1. Hidden States (S)**
- These are the **actual states of the system**, but they **cannot be observed directly**.
- The system moves between these states over time.
- **Example:** Weather = {Rainy, Cloudy, Sunny}.

# Components of a Hidden Markov Model

- **2. Observations (O)**
- These are the **visible outputs** generated by the hidden states.
- We can observe these outputs directly.
- **Example:** Emotions = {Happy, Neutral, Sad}.

# Components of a Hidden Markov Model

- **3. Transition Probability Matrix (A)**
- Defines the probability of moving from one hidden state to another.
- It answers the question:
- What is the probability that the next hidden state will be a particular state?
- Mathematically:
- $A = P(S_t \mid S_{t-1})$

# Components of a Hidden Markov Model

- 4. Emission Probability Matrix (B)
- Defines the probability of observing an output from a hidden state.
- It answers:
- **If the system is in a particular hidden state, what is the probability of observing a specific output?**
- Mathematically:
- $B = P(O_t \mid S_t)$

# Components of a Hidden Markov Model

- **5. Initial State Probability (π)**
- Specifies the probability of the system starting in each hidden state.
- Mathematically:
- $\pi = P(S_1)$

# When is HMM Used?

- Use HMM when:
- The data is **sequential** (events occur over time).
- The **true states are hidden** or unknown.
- Only the **observations** are available.
- The current state depends only on the **previous state** (Markov assumption).

# Common Applications of HMM

- **1. Speech Recognition**
- **Hidden States:** Spoken words or phonemes.
- **Observations:** Audio signals.
- **Purpose:** Convert speech into text.

# Common Applications of HMM

- **Weather Prediction**
- **Hidden States:** Weather (Sunny, Rainy, Cloudy).
- **Observations:** People's activities, umbrella usage, emotions.
- **Purpose:** Estimate the actual weather from observations.

# Common Applications of HMM

- **3. Part-of-Speech (POS) Tagging**
- **Hidden States:** Grammatical tags (Noun, Verb, Adjective).
- **Observations:** Words in a sentence.
- **Purpose:** Identify the grammatical role of each word.
- **Example:**
  Sentence: *Students study AI*.
- Hidden tags:
- Students → Noun
- Study → Verb
- AI → Noun

# Common Applications of HMM

- **4. Handwriting Recognition**
- **Hidden States:** Characters or letters.
- **Observations:** Pen strokes or images.
- **Purpose:** Convert handwritten text into digital text.

# Question 1

- A Hidden Markov Model has two hidden states:
- Rainy (R)
- Sunny (S)
- **Initial Probabilities**
- $\pi = [0.6,\ 0.4]$

| From/To | Rainy | Sunny |
|---|---|---|
| Rainy | 0.7 | 0.3 |
| Sunny | 0.4 | 0.6 |

# Question 1

**Emission Matrix**

| State | Umbrella | No Umbrella |
|---|---|---|
| Rainy | 0.9 | 0.1 |
| Sunny | 0.2 | 0.8 |

**Question**

If the first observation is **Umbrella**, calculate the probability that the observation was generated from each hidden state.

# Answer

- For Rainy
- $P(R, U) = 0.6 \times 0.9 = 0.54$
- For Sunny
- $P(S, U) = 0.4 \times 0.2 = 0.08$
- **Final Answer**
- Rainy = **0.54**
- Sunny = **0.08**
- Since **0.54 > 0.08**, the observation is most likely generated from the **Rainy** state.

# **Forward Algorithm** in a Hidden Markov Model.

- **Initialization**
- $\boxed{\alpha_1(i) = \pi(i) \times b_i(O_1)}$
- **Recursion**
- $\boxed{\alpha_t(j) = \left[\sum_i \alpha_{t-1}(i) a_{ij}\right] \times b_j(O_t)}$
- **Termination**
- $\boxed{P(O) = \sum_j \alpha_T(j)}$

Where:
•$\alpha_t(j)$= Forward probability of being in state $j$at time $t$
•$a_{ij}$= Transition probability from state $i$to state $j$
•$b_j(O_t)$= Emission probability of observation $O_t$from state $j$

# **Forward Algorithm** in a Hidden Markov Model. Exampl2

**Given**

**Hidden States**

•Rainy (R)

•Sunny (S)

**Initial Probabilities**

$$\pi(R) = 0.6, \pi(S) = 0.4$$

**Transition Matrix**

| From | R | S |
|---|---|---|
| R | 0.7 | 0.3 |
| S | 0.4 | 0.6 |

# **Forward Algorithm** in a Hidden Markov Model. Exampl2

**Emission Matrix**

**Observation Sequence**

$$O = (U, mbrella\ No\ Umbrella)$$

| State | Umbrella | No Umbrella |
|---|---|---|
| R | 0.9 | 0.1 |
| S | 0.2 | 0.8 |

# Step 1: Initialization (Day 1)

- The initialization formula is:
- $\boxed{\alpha_1(i) = \pi(i) \times b_i(O_1)}$

**For Rainy (R)**

$$\alpha_1(R) = \pi(R) \times b_R(Umbrella)$$
$$= 0.6 \times 0.9$$
$$= 0.54$$

**For Sunny (S)**

$$\alpha_1(S) = \pi(S) \times b_S(Umbrella)$$
$$= 0.4 \times 0.2$$
$$= 0.08$$

**Therefore,**

$$\boxed{\alpha_1(R) = 0.54, \alpha_1(S) = 0.08}$$

# Step 2: Recursion (Day 2)

- Observation = **No Umbrella**
- **Calculate Forward Probability for Rainy**
- Using the Forward Algorithm formula:
- $\alpha_2(R) = [\alpha_1(R)a_{RR} + \alpha_1(S)a_{SR}] \times b_R(No\ Umbrella)$
- Substitute the values:
- $= [(0.54 \times 0.7) + (0.08 \times 0.4)] \times 0.1$
- $= (0.378 + 0.032) \times 0.1$
- $= 0.410 \times 0.1$
- $= 0.041$
- Therefore,
- $\boxed{\alpha_2(R) = 0.041}$

# Calculate Forward Probability for Sunny

- Using the formula:
- $\alpha_2(S) = [\alpha_1(R)a_{RS} + \alpha_1(S)a_{SS}] \times b_S(No\ Umbrella)$
- Substitute the values:
- $= [(0.54 \times 0.3) + (0.08 \times 0.6)] \times 0.8$
- $= (0.162 + 0.048) \times 0.8$
- $= 0.210 \times 0.8$
- $= 0.168$
- Therefore,
- $\boxed{\alpha_2(S) = 0.168}$

# Step 3: Termination

- The probability of the observation sequence is the sum of the forward probabilities at the last time step.
- Formula:
- $\boxed{P(O) = \sum_j \alpha_T\,(j)}$
- Substitute the values:
- $P(O) = \alpha_2(R) + \alpha_2(S)$
- $= 0.041 + 0.168$
- $= 0.209$